

Overview of Image segmentation

Subject: Image segmentation

1.

Multi-scale segmentation Image segmentations are computed at multiple scales in scale space and sometimes propagated from coarse to fine scales; see scale-space segmentation. Segmentation criteria can be arbitrarily complex and may take into account global as well as local criteria. A common requirement is that each region must be connected in some sense.

2.

the algorithm selects the newly formed graph. Simulated annealing requires the input of temperature schedules which directly affects the speed of convergence of the system, as well as energy threshold for minimization to occur.

3.

Isolated Point Detection The detection of isolated points in an image is a fundamental part of image segmentation. This process primarily depends on the second derivative, indicating the use of the Laplacian operator. The Laplacian of a function $f(x, y)$ is given by:

4.

$$i = \begin{cases} i_{\text{new}}, & \text{if } \Delta U \leq 0, \\ i_{\text{new}}, & \text{if } \Delta U > 0 \text{ and } \delta < e^{-\Delta U / T}, \\ i_{\text{old}}, & \text{otherwise} \end{cases}$$

5.

Parametric methods Lagrangian techniques are based on parameterizing the contour according to some sampling strategy and then evolving each element according to image and internal terms. Such techniques are fast and efficient, however the original "purely parametric" formulation (due to Kass, Witkin and Terzopoulos in 1987 and known as "snakes"), is generally criticized for its limitations regarding the choice of sampling strategy, the internal geometric properties of the curve, topology changes (curve splitting and merging), addressing problems in higher dimensions, etc.. Nowadays, efficient "discretized" formulations have been developed to address these limitations while maintaining high efficiency. In both cases, energy minimization is generally conducted using a steepest-gradient descent, whereby derivatives are computed using, e.g., finite differences.

6.

Level-set methods The level-set method was initially proposed to track moving interfaces by Dervieux and Thomasset in 1979 and 1981 and was later reinvented by Osher and Sethian in 1988. This has spread across various imaging domains in the late 1990s. It can be used to efficiently address the problem of curve/surface/etc. propagation in an implicit manner. The central idea is to represent the evolving contour using a signed function whose zero corresponds to the actual contour. Then, according to the motion equation of the contour, one can easily derive a similar flow for the implicit

surface that when applied to the zero level will reflect the propagation of the contour. The level-set method affords numerous advantages: it is implicit, is parameter-free, provides a direct way to estimate the geometric properties of the evolving structure, allows for change of topology, and is intrinsic. It can be used to define an optimization framework, as proposed by Zhao, Merriman and Osher in 1996. One can conclude that it is a very convenient framework for addressing numerous applications of computer vision and medical image analysis. Research into various level-set data structures has led to very efficient implementations of this method.

7.

One-dimensional hierarchical signal segmentation Witkin's seminal work in scale space included the notion that a one-dimensional signal could be unambiguously segmented into regions, with one scale parameter controlling the scale of segmentation. A key observation is that the zero-crossings of the second derivatives (minima and maxima of the first derivative or slope) of multi-scale-smoothed versions of a signal form a nesting tree, which defines hierarchical relations between segments at different scales. Specifically, slope extrema at coarse scales can be traced back to corresponding features at fine scales. When a slope maximum and slope minimum annihilate each other at a larger scale, the three segments that they separated merge into one segment, thus defining the hierarchy of segments.

8.

Variational methods The goal of variational methods is to find a segmentation which is optimal with respect to a specific energy functional. The functionals consist of a data fitting term and a regularizing terms. A classical representative is the Potts model defined for an image f by

9.

$$\operatorname{argmin}_{u, K} \gamma |K| + \mu \int_K C |\nabla u|^2 dx + \int (u - f)^2 dx.$$

10.

Application of Isolated Point Detection in X-ray Image Processing The detection of isolated points has significant applications in various fields, including X-ray image processing. For instance, an original X-ray image of a turbine blade can be examined pixel-by-pixel to detect porosity in the upper-right quadrant of the blade. The result of applying an edge detector's response to this X-ray image can be approximated. This demonstrates the segmentation of isolated points in an image with the aid of single-pixel probes.

11.

Pick K cluster centers, either randomly or based on some heuristic method, for example K-means++ Assign each pixel in the image to the cluster that minimizes the distance between the pixel and the cluster center Re-compute the cluster centers by averaging all of the pixels in the cluster Repeat steps 2 and 3 until convergence is attained (i.e. no pixels change clusters) In this case, distance is the squared or absolute difference between a pixel and a cluster center. The difference is typically based on pixel color, intensity, texture, and location, or a weighted combination of these factors. K can be selected manually, randomly, or by a heuristic. This algorithm is guaranteed to converge, but it may not return

the optimal solution. The quality of the solution depends on the initial set of clusters and the value of K . The Mean Shift algorithm is a technique that is used to partition an image into an unknown apriori number of clusters. This has the advantage of not having to start with an initial guess of such parameter which makes it a better general solution for more diverse cases.

12.

The boundary encoding leverages the fact that regions in natural images tend to have a smooth contour. This prior is used by Huffman coding to encode the difference chain code of the contours in an image. Thus, the smoother a boundary is, the shorter coding length it attains. Texture is encoded by lossy compression in a way similar to minimum description length (MDL) principle, but here the length of the data given the model is approximated by the number of samples times the entropy of the model. The texture in each region is modeled by a multivariate normal distribution whose entropy has a closed form expression. An interesting property of this model is that the estimated entropy bounds the true entropy of the data from above. This is because among all distributions with a given mean and covariance, normal distribution has the largest entropy. Thus, the true coding length cannot be more than what the algorithm tries to minimize. For any given segmentation of an image, this scheme yields the number of bits required to encode that image based on the given segmentation. Thus, among all possible segmentations of an image, the goal is to find the segmentation which produces the shortest coding length. This can be achieved by a simple agglomerative clustering method. The distortion in the lossy compression determines the coarseness of the segmentation and its optimal value may differ for each image. This parameter can be estimated heuristically from the contrast of textures in an image. For example, when the textures in an image are similar, such as in camouflage images, stronger sensitivity and thus lower quantization is required.